

AMIT KUMAR MOHANTY

Machine Learning Engineer

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EDUCATION

Master of Technology
Artificial Intelligence and Machine Learning
BITS Pilani
📅 2026 - 2028 (Pursuing)

Bachelor of Technology
Computer Science & Engineering (AI & ML)
K.R. Mangalam University
📅 2021 - 2025 | GPA 8.5/10

SKILLS

Languages

- Python • C++ • SQL • R

ML & Frameworks

- TensorFlow • PyTorch • Scikit-learn • OpenCV • FastAPI

AI / GenAI

- LLMs • Agentic AI • RAG • Transformers • NLP
• Prompt Engineering • Hugging Face • BERT

Agent Frameworks

- LangChain • LangGraph • LangSmith

MLOps & Deployment

- Docker • Kubernetes • CI/CD • MLflow • Model Serving • Git • FAISS/Chroma

Cloud & Data

- AWS • GCP (Vertex AI) • MySQL • MongoDB • Pandas • NumPy • Power BI

LANGUAGES

- English • Hindi • German (A1)

INTERESTS

- Tennis • Running • Singing • Sketching • Reading • Philosophy • Travelling • Learning New Technologies

PROFESSIONAL SUMMARY

Machine Learning Engineer with 1+ year of experience architecting production-grade AI systems, high-throughput model-serving platforms, and event-driven applications. Proven track record of leading technical initiatives, designing low-latency services at enterprise scale, and delivering measurable business impact through ML engineering and AI-driven automation. Outside of work, a competitive tennis player and avid sketch artist, pursuits that sharpen discipline, focus, and creativity.

WORK EXPERIENCE

Research Associate – AI

Keywords Studios India

📅 May 2025 - Present 📍 Gurugram, Haryana

- Awarded "Rookie of the Year 2025" for surpassing all performance KPIs through advanced model training and data-pipeline optimization.
- Trained and optimized enterprise-grade Agentic AI models to autonomously execute complex multi-step workflows (flight booking, e-commerce ordering, web data retrieval), improving task-completion reliability.
- Trained and refined an agentic browser-automation AI that autonomously navigates websites and completes real-world tasks on a user's behalf, delivering client-ready model iterations.
- Led quality-control and evaluation of the agentic browser-automation system with a global team, running structured test reviews and daily syncs to validate task accuracy before client delivery.
- Designed and orchestrated multi-agent workflows with LangChain and LangGraph, enabling autonomous planning, tool use, and self-correction across chained tasks.
- Built custom data pipelines and curated high-quality instruction datasets, delivering 85% of self-generated training data to production as the first member of a global team.

Artificial Intelligence Intern

Infosys

📅 Nov 2024 - Feb 2025 📍 Remote

- Engineered an AI call-center system with LangChain and LangSmith, boosting model response accuracy by 35% and elevating customer-AI interaction quality.
- Gained deep, hands-on expertise in LangChain and LangGraph to design, orchestrate, and deploy production LLM workflows, strengthening end-to-end deployment quality.
- Developed and deployed LLM features that reduced system latency by 25%, streamlining the model deployment process.
- Designed and A/B-tested prompt-engineering strategies with automated LangSmith evaluation, reducing hallucinations by 20%.

PROJECTS

Dynamic Screen Companion

- Built a real-time multi-modal assistant integrating the Gemini API and a high-performance FastAPI backend; optimized OCR pipelines to 95%+ accuracy, reducing manual task time by 40%.
- Implemented concurrent inference streaming with live voice feedback for smooth, hands-free assistance at low latency.

Diabetic Optiscan

- Led a Vision Transformer (ViT-B/16) model on the APTOS dataset with a custom wavelet technique that boosted detection precision by 50%; co-authored a peer-reviewed paper.
- Applied image preprocessing and augmentation on high-resolution retinal scans to improve model generalization and diagnostic consistency.

Currency Detection App for Visually Impaired

- Trained an image-classification model on 200,000+ banknote images achieving 96% accuracy, deployed in a mobile app with voice feedback to help prevent fraud.
- Optimized the model for on-device inference with TensorFlow Lite, enabling fast, offline real-time recognition on low-end smartphones.

CERTIFICATIONS

Data Science Professional – Ducat India

📅 Apr 2024 - Apr 2025

- Attended an in-person, project-based program building strong hands-on foundations in machine learning, data analysis, and deep learning.

AI/ML for Geodata Analysis – ISRO

- Applied AI/ML to satellite imagery and geospatial datasets for terrain classification and remote-sensing analysis.